परीक्षा प्रश्न पत्र/**Examination Question Paper**

PGCP-ITISS August 2026

परीक्षा का प्रकार / **EXAM Type:** मुख्य / ~~पुनः परीक्षा~~ **Main /~~Re-Exam~~**

मोड्यूल का नाम **Module Name: ITIM and DevOps**

समूह /**SET: B**

दिनांक/**DATE:12/11/2026** कालावधि/**Duration: 2 Hrs.**

अधिकतम अंक/**Max. Marks: 40**

नाम/**Name:** Soumil Sathe

पीआरएन/**PRN:** 260840123048

निर्देश / ***Instructions:***

1. All questions are compulsory.

2. Please create a PDF file that includes the code/steps and corresponding screenshots and outputs for each question, organized in numerical order

3. Attach screenshots for every step and highlight the steps in that screenshot.

4. Please make sure to close all required files, including PDFs, before creating a zip file to prevent corruption of the zip file

5. Q1, Q2 & Q3 as folders and keep these folders in a single folder and zip.

6. Zip File should be renamed with FullPRN_ModuleName_Date of Lab Exam. (For example, 260840123001_* ITIM and DevOps*_12/11/2026.zip)

- ***Upload using Link- https://actslms.cdac.in***
- ***UserID- Full PRN***
- ***Password- FirstName@Full PRN***

Q1. Attach Screenshots and highlight each and every step **[20 Marks]**

a) Create a VPC named "SecureVPC_" with CIDR 192.168.0.0/16. [5 Marks]

b) Create one public and one private subnet. Launch: [10 Marks]

a. Public EC2 instance in the public subnet.
b. Private EC2 instance in the private subnet.

i. Security Rules:
   1. Public EC2: SSH from your system only.
   2. Private EC2: SSH only from Public EC2 Security Group.
   3. Ensure the private instance does not receive a public IP.

c) Create a file named <PRN>.txt in the Public EC2 instance containing 10 lines of text. Transfer the file to the Private EC2 instance and display the first five lines using Linux commands. [5 Marks]

OR

Q1. Multi-Deployment Application using Kubernetes **[20 Marks]**

a) Create a Deployment named frontend_<PRN> using the nginx image with 2 replicas. [4 Marks]

b) Create a Deployment named backend_<PRN> using the httpd image with 2 replicas. [4 Marks]

c) Expose the frontend Deployment using a NodePort service and access it through a browser. [4 Marks]

d) Expose the backend Deployment using a ClusterIP service. [3 Marks]

e) Create a temporary Pod and verify connectivity to the backend using its ClusterIP service name. [3 Marks]

f) Scale the backend Deployment to 3 replicas and verify the updated Pod count. [2 Marks]

Q2. Attach Screenshots and highlight each and every step **[10 Marks]**

a) Pull the ubuntu:22.04 image and run a container.

b) Create a Docker Volume named backup_data and mount it to /data. show the docker volumes

c) Create <PRN.txt> file inside /data.

d) Stop and remove the container. Create a new container using the same volume and verify the files exist inside the container. Show the content of file from the container

e) Install curl and nano inside the container, and create a new image from the running container and tagged as "<prn>:v2". show the docker images

Q3. Create a VM instance of your choice on Oracle Virtual Box or on VMWare Virtual Box and do the following and Capture logs/Snapshots of each step **[10 Marks]**

a) Initialize the local git repository.

b) Create a <PRN>.txt file with "PRN - <PRN>" and commit to main branch with commit message "first commit"

c) Add your name to the new line in <PRN>.txt and commit with message "second commit"

d) Add the new "welcome to Ditiss" in <PRN>.txt and commit with "third commit"

e) Restore git to new First commit

**********शुभकामनाएं/ALL THE BEST! **********